

Describe how data validation and data verification help protect the integrity of data

9608_s21_qp_13_sc_1.b.i

QUESTION

(b) (i) Validation is one way to protect the integrity of input data.

Identify **one other** method to protect the integrity of input data.

..... [1]

MARK SCHEME

1(b)(i)	Verification // by example	1
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9608_s21_qp_11_sc_6.e

QUESTION

(e) Examine the following list of terms:

bandwidth	browser	compiler	HTML	interpreter
JavaScript	PHP	router	server	server-side
transmission	validation			

Complete the following sentences by filling in the missing terms from the list.
Some terms may be used more than once. Some terms are not used.

..... can be performed both client-side and server-side.

It is performed more rapidly by the because there is no delay in transmitting and receiving data to and from the

It is also performed on the server-side, because the client's may not support, so the data will still need checking to avoid errors.

[5]

MARK SCHEME

6(e)	1 mark for each correctly completed term Validation can be performed both client-side and server-side. It is performed more rapidly by the browser because there is no delay in transmitting and receiving data, to and from the server . It is also performed server-side because the client's browser may not support JavaScript , so the data will still need checking to avoid errors.	5
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9608_s20_qp_11_sc_1.a

QUESTION

1 A computer program makes use of data validation routines and verification of data input. (a) Complete these two sentences about data validation and verification. 1. checks that the data entered is reasonable. One example is 2. checks that the data entered is the same as the original. One example is [4]

MARK SCHEME

1(a)	1 mark for each correctly completed term. Validation checks that the data entered is reasonable. One example is a presence check . Verification checks that the data entered is the same as the original. One example is double entry .	4
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QUESTION

- (ii) The supermarket is concerned about the security and integrity of the data on the server.

Identify **two** methods that can be used to minimise the security risk to the data, and **one** method to protect the integrity of the data.

Security 1

.....

Security 2

.....

Integrity

.....

[3]

MARK SCHEME

1(c)(ii)	1 mark for each security method to max 2, 1 mark for integrity Security encryption access rights username and password // biometrics // user accounts backup // disk mirroring firewall Physical methods (e.g. CCTV, locked rooms etc.) Integrity checksum parity validation on input	3
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QUESTION

(c) Four error detection measures are shown.

Draw **one** line from each error detection measure to indicate whether it is verification or validation.

Error detection measure

Type check

Proof reading

Check digit

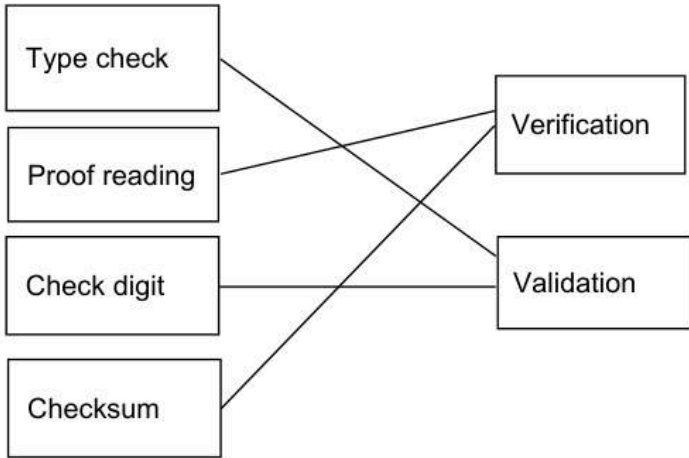
Checksum

Verification

Validation

[4]

MARK SCHEME

4(c)	1 mark for each correct line Error detection measure  The diagram shows four boxes on the left: 'Type check', 'Proof reading', 'Check digit', and 'Checksum'. On the right are two boxes: 'Verification' and 'Validation'. Lines connect 'Type check' to 'Verification', 'Proof reading' to 'Verification', 'Check digit' to 'Validation', and 'Checksum' to 'Validation'.	4
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QUESTION

(c) A task for staff at the start of the school year is to key in new pupil data from a paper document.

The data is entered to a screen form and includes the data verification of some fields.

Describe what is meant by **verification**.

.....

.....

.....

.....[2]

MARK SCHEME

3(c)	1 Mark per bullet, max 2 Checking that the data entered matches / is consistent with that of the source. Comparison of two versions of the data Examples include double entry, visual checking, proof reading etc... In the event of a mismatch – the user is forced to re-enter the data By example, e.g. creation of a password Does not check data is sensible/acceptable	2
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QUESTION

(iv) Describe the purpose of the validation check that the code performs.

.....

.....[1]

MARK SCHEME

7(c)(iv)	Checks that the product code has not be left blank // presence check on product code	1
----------	--	----------

QUESTION

- (ii) *"Data validation is used to make sure that data keyed in are the same as the original data supplied."*

.....
.....
.....
.....[2]

MARK SCHEME

- (ii) Any **two** from:

- This is an explanation of data verification (not validation)
- Data validation ensures that data is reasonable / sensible / within a given criteria
- Original data may have been entered correctly but is not reasonable (e.g. age of 210)

[2]

QUESTION

- (ii) *"Data validation is used to make sure that data keyed in are the same as the original data supplied."*

.....
.....
.....
.....[2]

MARK SCHEME

- (ii) Any **two** from:

- This is an explanation of data verification (not validation)
- Data validation ensures that data is reasonable / sensible / within a given criteria
- Original data may have been entered correctly but is not reasonable (e.g. age of 210)

[2]

QUESTION

9 (a) Give a brief description of each of the following terms:

Validation
.....
.....
Verification
.....
.....[2]

MARK SCHEME

9 (a) One mark for validation, one mark for verification.

Validation

- check whether data is reasonable / meets given criteria

Verification

- method to ensure data which is copied / transferred is the same as the original
- entering data twice and computer checks both sets of data
- check entered data against original document / source

[2]

QUESTION

9 (a) Give a brief description of each of the following terms:

Validation
.....
.....
Verification
.....
.....[2]

MARK SCHEME

9 (a) One mark for validation, one mark for verification.

validation

- check whether data is reasonable / meets given criteria

verification

- method to ensure data which is copied / transferred is the same as the original
- entering data twice and computer checks both sets of data
- check entered data against original document / source

[2]

9608_s15_qp_13_sc_3.c.i

QUESTION

(c) Data integrity is required at the input stage and also during transfer of the data.

(i) State **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer.

.....

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

(c) (i) **one mark** for each way of maintaining data security + **one mark** for an example/enhancement

- validation (to ensure data is reasonable)
- examples include range checks, type checks, length checks, ...
- verification (checks if data input matches original/if transmitted data matches original)
- can use double data entry or visual check/other methods such as parity checks
- doesn't check whether or not data is reasonable

[3]

QUESTION

- (ii) State **two** ways of maintaining data integrity during data transmission. Use examples to help explain your answer.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

- (ii) **one mark** for each way of maintaining data integrity + **one mark** for an example/enhancement

- parity checking
- one of the bits is reserved as parity bit
- e.g. 1 0 1 1 0 1 1 0 uses odd parity
- number of 1s must be odd
- parity is checked at receiver's end
- a change in parity indicates data corruption

- check sum
- adds up bytes in data being sent and sends check sum with the data
- calculation is re-done at receiver's end
- if not the same sum then the data has been corrupted during transmission [3]

QUESTION

- (b) Sensors are used to monitor seismic activity. At the end of each day, all the data are transmitted to a central computer. This is hundreds of kilometres away.

Describe **one** way of ensuring that the integrity of the data is retained during the transmission stage.

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- (b) 1 mark for name + 3 marks for description

parity check

- uses even or odd parity which is decided before data sent
- each byte has a parity bit
- parity bit is set to 0 or 1 to make parity for byte correct
- after transmission, parity of each byte re-checked
- if it is different, then an error is flagged
- any reference to use of parity blocks/parity byte to (identify position of incorrect bit)

checksum

- a calculation is carried out on the data to be sent (checksum)
- the result is sent, along with data to recipient
- checksum is re-calculated at receiving end
- if both sums are the same, no error has occurred
- if the sums are different, the data has been corrupted during transmission
- request is sent to re-send data

[4]

Describe and use methods of data validation

9608_s21_qp_13_sc_1.a

QUESTION

1 An adventure sports company has a website.

(a) Customers can book courses using an online booking form. The booking form contains a number of fields.

The following table gives a description of the validation for each field.

Write the validation type for each validation description in the table.

Field	Validation description	Validation type
Name	A name must be entered	
Date of Birth	Entered as dd/mm/yyyy	
Telephone Number	A limit of 15 characters can be entered	
Experience Level	Only values between 1 and 5 can be entered	

[4]

MARK SCHEME

1(a)	1 mark for each validation type	4															
	<table><tr><th>Field</th><th>Validation description</th><th>Validation type</th></tr><tr><td>Name</td><td>A name must be entered</td><td>Presence check</td></tr><tr><td>Date of birth</td><td>Entered as dd/mm/yyyy</td><td>Format check</td></tr><tr><td>Telephone number</td><td>A limit of 15 characters can be entered</td><td>Length check</td></tr><tr><td>Experience level</td><td>Only values between 1 and 5 can be entered</td><td>Range check</td></tr></table>	Field	Validation description	Validation type	Name	A name must be entered	Presence check	Date of birth	Entered as dd/mm/yyyy	Format check	Telephone number	A limit of 15 characters can be entered	Length check	Experience level	Only values between 1 and 5 can be entered	Range check	
Field	Validation description	Validation type															
Name	A name must be entered	Presence check															
Date of birth	Entered as dd/mm/yyyy	Format check															
Telephone number	A limit of 15 characters can be entered	Length check															
Experience level	Only values between 1 and 5 can be entered	Range check															

9608_s21_qp_12_sc_5.e.i

QUESTION

(e) Data validation has been used in line 15 of the JavaScript code.

(i) Identify the type of data validation used in line 15.

..... [1]

MARK SCHEME

5(e)(i)	Presence check	1
---------	----------------	---

9608_s21_qp_12_sc_5.e.ii

QUESTION

(ii) Identify **two** other data validation checks that could be used.

1

2 [2]

MARK SCHEME

5(e)(ii)	1 mark for each (max 2) • range check • length check • type check	2
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9608_s20_qp_11_sc_1.a

QUESTION

1 A computer program makes use of data validation routines and verification of data input.

(a) Complete these **two** sentences about data validation and verification.

1. checks that the data entered is reasonable. One example is

2. checks that the data entered is the same as the original. One example is

[4]

MARK SCHEME

1(a)	1 mark for each correctly completed term. Validation checks that the data entered is reasonable. One example is a presence check. Verification checks that the data entered is the same as the original. One example is double entry.	4
------	---	----------

9608_s18_qp_13_sc_4.c

QUESTION

(c) Four error detection measures are shown.

Draw **one** line from each error detection measure to indicate whether it is verification or validation.

Error detection measure

Type check

Proof reading

Check digit

Checksum

Verification

Validation

[4]

MARK SCHEME

4(c)	1 mark for each correct line Error detection measure	4
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9608_s18_qp_12_sc_3.c

QUESTION

(c) The following table shows five error detection measures.

Put **one** tick (✓) in each row to indicate whether the measure is validation or verification.

Measure	Validation	Verification
Checksum		
Format check		
Range check		
Double entry		
Check digit		

[5]

MARK SCHEME

3(c)	1 mark per each correct row	5																		
<table border="1"> <thead> <tr> <th>Measure</th><th>Validation</th><th>Verification</th></tr> </thead> <tbody> <tr> <td>Checksum</td><td></td><td>✓</td></tr> <tr> <td>Format check</td><td>✓</td><td></td></tr> <tr> <td>Range check</td><td>✓</td><td></td></tr> <tr> <td>Double entry</td><td></td><td>✓</td></tr> <tr> <td>Check digit</td><td>✓</td><td></td></tr> </tbody> </table>			Measure	Validation	Verification	Checksum		✓	Format check	✓		Range check	✓		Double entry		✓	Check digit	✓	
Measure	Validation	Verification																		
Checksum		✓																		
Format check	✓																			
Range check	✓																			
Double entry		✓																		
Check digit	✓																			

9608_w17_qp_13_sc_3.b.i

QUESTION

(b) A frequent task for staff is to key in new patient data from a paper document. The document includes the patient's personal ID number.

- (i) The Patient ID is a seven digit number. The database designer decides to use a check digit to verify each foreign key value that a user keys in for a Patient ID.

When a user assigns a primary key value to a Patient ID, the DBMS adds a modulus-11 check digit as an eighth digit. The DBMS uses the weightings 6, 5, 4, 3, 2 and 1 for calculating the check digit. It uses 6 as the multiplier for the most significant (leftmost) digit.

Show the calculation of the check digit for the Patient ID with the first six digits 786531.

Complete Patient ID[4]

MARK SCHEME

3(b)(i)

4

7	X	6	=	42
8	X	5	=	40
6	X	4	=	24
5	X	3	=	15
3	X	2	=	6
1	X	1	=	1
			Total:	128 / 11
				11 R 7
			Check digit:	11 - 7 = 4

1 mark for 6 values

1 mark for 2 steps
Accept 128 MOD 11 = 7

1 mark for subtraction

Answer: 786531 4 (1 mark for answer)

QUESTION

(ii) Name and describe **two** validation checks that the DBMS could carry out on each primary key value that a user keys in for a Patient ID.

- 1 Validation check
Description
.....
- 2 Validation check
Description
.....
- [4]

MARK SCHEME

3(b)(ii)	One mark for name of check One mark for description Max two checks Uniqueness check Each PatientID must be unique Length check Each PatientID is exactly 7 characters Format check / Type check All 7 characters must be <u>digits</u> Presence check PatientID must be entered	Max 4
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QUESTION

(iv) Describe the purpose of the validation check that the code performs.

.....
.....[1]

MARK SCHEME

7(c)(iv)	Checks that the product code has not be left blank // presence check on product code	1
----------	--	----------

QUESTION

- (v) Name and describe **two** other types of validation check that could be appropriate for this data capture form.

Validation check:

Description

.....

Validation check:

Description

.....

[4]

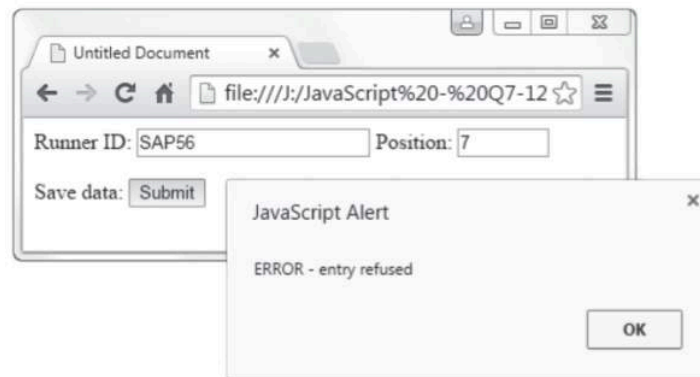
MARK SCHEME

7(c)(v)	Two checks from: One mark for check and one mark for description	Max 4
	Range check	1
	Check the number entered is (say) between 1 and 100	1
	Format check	1
	Checks the product code is a particular format // Checks the number has digit characters only // by example	1
	Length check	1
	The number of items has exactly five characters	1
	Existence check	1
	To ensure the product code has been assigned	1

QUESTION

(c) The following web page used for data capture consists of:

- two text boxes for the entry of:
 - a race runner's ID code
 - their finishing position in a race.
- a button that the user clicks to submit this runner's result.



```

1  <html>
2  <head>
3  <title>Untitled Document</title>
4  <script language="JavaScript">
5
6  function myButton_onmousedown()
7  {
8  var Output1 = "Runner ID OK";
9  var Output2 = "ERROR - entry refused";
10
11 var Runner ID = document.forms["form1"]["txtRunnerID"].value;
12 //           || in Javascript is the 'OR' operator
13 if (RunnerID.substr(0,3) == "VAR" || RunnerID.substr(0,3) == "CAM")
14 {
15     alert(Output1)
16 }
17 else
18 {
19     alert(Output2)
20 }
21 }
22 </script>
23
24 </head>
25 <body>
26 <form name = form1>
27   <label>Runner ID: </label>
28   <input type="text" name="txtRunnerID" >
29   <label>Position: </label>
30   <input type="text" name="txtPosition" size = "5" >
31   <p>
32     <label>Save data: </label>
33     <input type="button" name="btnSubmit" Value = "Submit"
34
35     onMouseDown = "myButton_onmousedown()" >
36   </p>
37 </form>
38
39 </body>
40 </html>

```

- (i) The developer has used three variables in the JavaScript code. State the identifiers used.

1.
2.
3.

[2]

MARK SCHEME

6(c)(i)	Output1, Output2 RunnerID // Runner ID	1 1	2
---------	---	--------	---

9608_s17_qp_12_sc_6.c.ii

QUESTION

- (ii) The button has an event whose identifier is `onMouseDown`. When the mouse button is clicked, some code is run.

State the line numbers which contain this code.

From line to line

[1]

MARK SCHEME

6(c)(ii)	6 – 21		1
----------	--------	--	---

9608_s17_qp_12_sc_6.c.iii

QUESTION

- (iii) The JavaScript code uses a selection statement.

State the line number which contains its condition.

Line number:

[1]

MARK SCHEME

6(c)(iii)	13		1
-----------	----	--	---

9608_s17_qp_12_sc_6.c.iv

QUESTION

- (iv) Describe the purpose of the validation check that the code performs.

.....
.....[1]

MARK SCHEME

6(c)(iv)	Checks that the <code>RunnerID</code> entered starts with the characters CAM or VAR only	1
----------	--	---

9608_s17_qp_12_sc_6.c.v

QUESTION

- (v) Name and describe **two** other types of validation check which could be appropriate for this data capture form.

Validation check:

Description

.....

Validation check:

Description

.....[4]

MARK SCHEME

6(c)(v)	Two checks from: One mark for check and one mark for description Format check RunnerID is three letter characters followed by two digit characters //Position is digit characters only Length check RunnerID has exactly five characters Range check The value for Position is between 1 and (say) 50 Presence check The text box for RunnerID or Position is not empty Existence check To ensure that RunnerID has been registered Uniqueness check To ensure no two runners have the same number	1 1 1 1 1 1 1 1	Max 4
---------	--	---	-------

9608_s17_qp_11_sc_7.c.iv

QUESTION

(iv) Describe the purpose of the validation check that the code performs.

.....
.....[1]

MARK SCHEME

7(c)(iv)	Checks that the product code has not be left blank // presence check on product code	1
----------	--	---

QUESTION

- (v) Name and describe **two** other types of validation check that could be appropriate for this data capture form.

Validation check:

Description

.....

Validation check:

Description

.....

[4]

MARK SCHEME

7(c)(v)	Two checks from: One mark for check and one mark for description	Max 4
	Range check	1
	Check the number entered is (say) between 1 and 100	1
	Format check	1
	Checks the product code is a particular format // Checks the number has digit characters only // by example	1
	Length check	1
	The number of items has exactly five characters	1
	Existence check	1
	To ensure the product code has been assigned	1

QUESTION

(c) Data integrity is required at the input stage and also during transfer of the data.

(i) State **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

(c) (i) **one mark** for each way of maintaining data security + **one mark** for an example/enhancement

- validation (to ensure data is reasonable)
- examples include range checks, type checks, length checks, ...
- verification (checks if data input matches original/if transmitted data matches original)
- can use double data entry or visual check/other methods such as parity checks
- doesn't check whether or not data is reasonable

[3]

Describe and use methods of data verification during data entry and data transfer

9608_s21_qp_13_sc_1.b.i

QUESTION

(b) (i) Validation is one way to protect the integrity of input data.

Identify **one other** method to protect the integrity of input data.

..... [1]

MARK SCHEME

1(b)(i)	Verification // by example	1
---------	----------------------------	----------

9608_s21_qp_13_sc_1.b.ii

QUESTION

(ii) The data input will be transferred to a central server.

Identify **two** measures to protect the integrity of the data during transfer.

1

.....

2

.....

[2]

MARK SCHEME

1(b)(ii)	1 mark per bullet point: • Parity • Checksum	2
----------	---	----------

QUESTION

- 1 A computer program makes use of data validation routines and verification of data input.
- (a) Complete these **two** sentences about data validation and verification.
1. checks that the data entered is reasonable. One example is
2. checks that the data entered is the same as the original. One example is
- [4]

MARK SCHEME

1(a)	1 mark for each correctly completed term. Validation checks that the data entered is reasonable. One example is a presence check. Verification checks that the data entered is the same as the original. One example is double entry.	4
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QUESTION

- (c) A computer uses parity blocks to check the data that has been received is the same as the data that has been transmitted.

The following is an example of a parity block.

	Parity bit	Data						
	1	1	1	1	0	0	0	1
	0	0	0	0	1	1	1	0
	1	1	0	1	1	0	0	1
Parity byte	1	1	0	1	1	0	0	1

- (i) Describe how a parity block check can identify a bit that has been corrupted during transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

2(c)(i)	1 mark per bullet to max 4 Each byte has a parity bit // horizontal parity An additional parity byte is sent with vertical (and horizontal) parity Each row and column must have an <u>even/odd number of 1s</u> Identify the incorrect row and column The intersection is the error	4
---------	---	----------

9608_s19_qp_13_sc_2.c.ii

QUESTION

- (ii) Give a situation where a parity block check **cannot** identify corrupted bits.

.....
 [1]

MARK SCHEME

2(c)(ii)	1 mark for correct answer Errors in an even number of bits (could cancel each other out)	1
----------	--	----------

9608_s18_qp_13_sc_4.a

QUESTION

- 4 Parity bits can be used to verify data.

- (a) The following binary number is transmitted using **odd** parity.

Add the missing parity bit.

**Parity
bit**

	0	1	1	1	0	1	0
--	---	---	---	---	---	---	---

[1]

MARK SCHEME

4(a)	Parity bit								1
	1	0	1	1	1	0	1	0	

QUESTION

- (b) In the following parity block, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

Circle the error in the data being transmitted.

	Parity bit	Data						
	0	0	1	1	0	1	0	1
	1	1	1	1	1	0	0	1
	1	0	1	0	1	0	0	0
	0	0	0	1	1	0	1	1
Parity byte	0	1	1	1	1	1	0	1

[1]

MARK SCHEME

4(b)

1 mark for correctly circled bit

1

Parity bit	Data							
0	0	1	1	0	1	0	1	
1	1	1	1	1	0	0	1	
1	0	1	0	1	0	0	0	
0	0	0	1	1	0	1	1	
Parity byte	0	1	1	1	1	1	0	1

QUESTION

(c) Four error detection measures are shown.

Draw **one** line from each error detection measure to indicate whether it is verification or validation.

Error detection measure

Type check

Proof reading

Check digit

Checksum

Verification

Validation

[4]

MARK SCHEME

4(c)	1 mark for each correct line Error detection measure <pre>graph LR; TC[Type check] --- V[Validation]; PR[Proof reading] --- Ver[Verification]; CD[Check digit] --- V; CS[Checksum] --- Ver;</pre>	4
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9608_s18_qp_12_sc_3.a

QUESTION

3 Parity bits can be used to verify data.

(a) The following binary number is transmitted using **odd** parity.

Add the missing parity bit.

**Parity
bit**

	0	1	0	0	0	0	0
--	---	---	---	---	---	---	---

[1]

MARK SCHEME

3(a)	1 mark for correct parity bit Parity bit <table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	0	0	1	0	0	0	0	0	1
0	0	1	0	0	0	0	0			

9608_s18_qp_12_sc_3.b

QUESTION

(b) In the following data transmitted, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

Circle the error in the data transmitted.

Parity bit	Data						
1	0	1	0	1	1	1	1
0	1	1	0	0	1	1	0
1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0
Parity byte	0	0	0	0	1	0	1

[1]

MARK SCHEME

3(b)

1 mark for the correct bit circled.

1

Parity bit	Data						
1	0	1	0	1	1	1	1
0	1	1	0	0	1	1	0
1	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0
Parity byte	0	0	0	0	1	0	1

9608_s18_qp_12_sc_3.c

QUESTION

(c) The following table shows five error detection measures.

Put **one** tick (✓) in each row to indicate whether the measure is validation or verification.

Measure	Validation	Verification
Checksum		
Format check		
Range check		
Double entry		
Check digit		

[5]

MARK SCHEME

3(c)	1 mark per each correct row	5																		
	<table> <tr> <th>Measure</th><th>Validation</th><th>Verification</th></tr> <tr> <td>Checksum</td><td></td><td>✓</td></tr> <tr> <td>Format check</td><td>✓</td><td></td></tr> <tr> <td>Range check</td><td>✓</td><td></td></tr> <tr> <td>Double entry</td><td></td><td>✓</td></tr> <tr> <td>Check digit</td><td>✓</td><td></td></tr> </table>	Measure	Validation	Verification	Checksum		✓	Format check	✓		Range check	✓		Double entry		✓	Check digit	✓		
Measure	Validation	Verification																		
Checksum		✓																		
Format check	✓																			
Range check	✓																			
Double entry		✓																		
Check digit	✓																			

9608_s18_qp_11_sc_6.a

QUESTION

6 Parity bits can be used to verify data.

(a) The following binary number is transmitted using **even** parity.

Add the missing parity bit.

**Parity
bit**

	1	0	1	1	0	1	0
--	---	---	---	---	---	---	---

[1]

MARK SCHEME

6(a)	1 mark for the correct parity Parity bit <table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table>	0	0	1	0	0	0	0	0	1
0	0	1	0	0	0	0	0			

9608_s18_qp_11_sc_6.b.i

QUESTION

(b) In the following parity block, the first column contains the parity bits, and the last row contains the parity byte. A device transmits the data using **even** parity.

(i) Circle the error in the data transmitted.

	Parity bit	Data						
	1	1	0	1	0	1	1	1
	1	0	0	0	1	1	1	0
	0	1	0	0	1	0	1	1
	1	1	1	0	1	1	1	1
Parity byte	1	1	1	1	1	0	0	1

[1]

MARK SCHEME

6(b)(i)	1 mark for the correct bit circled								1
		Parity bit	Data						
		1	1	0	1	0	1	1	1
		1	0	0	0	1	1	1	0
		0	1	0	0	1	0	1	1
		1	1	1	0	1	1	1	1
Parity byte	1	1	1	1	1	0	0	1	

9608_s18_qp_11_sc_6.b.ii

QUESTION

(ii) Explain how you identified the error.

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.....

.....

.....[2]

MARK SCHEME

6(b)(ii)	1 mark for each bullet point row and column has incorrect parity (odd instead of even) the intersection identifies the error	2
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9608_s18_qp_11_sc_6.c

QUESTION

(c) The data received can contain errors that are not detected using parity bits.

Explain how this can happen.

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.....

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.....[2]

MARK SCHEME

6(c)	1 mark per bullet to max 2 Errors in an even number of bits (in the same row or column) Prevents error being identified Could appear to be correct	2
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9608_s18_qp_11_sc_6.d

QUESTION

(d) Parity is not the only method to verify the data has been sent correctly.

Name **and** describe **one** other method of data verification during data transfer.

Name

Description

.....

.....

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.....

.....

.....[3]

MARK SCHEME

6(d)	1 mark for the name and max 2 marks for description Check sum A calculation is done on a block of data The result is transmitted with the data Calculation repeated at receiving end Results compared If different an error has occurred Hash total Total of several fields of data Including fields not usually used in calculations The result is transmitted with the data Calculation repeated at receiving end Results compared If different an error has occurred	3
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9608_w17_qp_12_sc_3.c

QUESTION

(c) A task for staff at the start of the school year is to key in new pupil data from a paper document.

The data is entered to a screen form and includes the data verification of some fields.

Describe what is meant by **verification**.

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.....[2]

MARK SCHEME

3(c)	1 Mark per bullet, max 2 Checking that the data entered matches / is consistent with that of the source. Comparison of two versions of the data Examples include double entry, visual checking, proof reading etc... In the event of a mismatch – the user is forced to re-enter the data By example, e.g. creation of a password Does not check data is sensible/acceptable	2
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QUESTION

- 5 A computer receives data from a remote data logger. Each data block is a group of 8 bytes. A block is made up of seven data bytes and a parity byte.
- Each data byte has a parity bit using odd parity. The parity byte also uses odd parity.
- The following table shows a data block before transmission. Bit position 0 is the parity bit.

Bit position							
7	6	5	4	3	2	1	0
1	1	0	0	1	1	0	1
0	0	1	0	0	0	0	0
1	0	0	1	1	1	0	A
1	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0
1	1	0	0	0	1	1	B
0	0	0	0	0	0	0	0

← Data bytes

0	1	1	0	1	1	0	0
---	---	---	---	---	---	---	---

← Parity byte

- (a) (i) Describe how the data logger calculates the parity bit for each of the bytes in the data block.
-
-
-[2]

MARK SCHEME

5(a)(i)	Count the number of one bits in the <u>first seven</u> bit positions	1	2
	Add a 0 or 1 to bit position 0, to make the count of one bits an <u>odd</u> number	1	

QUESTION

- (ii) State the two missing parity bits labelled A and B.
- A =
- B = [1]

MARK SCHEME

5(a)(ii)	A = 1 B = 1	1
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9608_s17_qp_13_sc_5.a.iii

QUESTION

- (iii) Describe how the computer uses the parity byte to perform a further check on the received data bytes.

.....

[2]

MARK SCHEME

5(a)(iii)	Two from: A parity bit is worked out for each <u>column</u> 1 The computer checks the parity of each bit position in parity byte // the computer generates copy of the parity byte and <u>compares</u> 1 If incorrect parity then there is an error in the data received // No parity error means no error in the data received 1 The position of the incorrect bit can be determined 1	2
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9608_s17_qp_13_sc_5.b.i

QUESTION

- (b) (i) A second data block is received as shown in the following table. There are errors in this data block.

Identify and then circle **two** bits in the table which must be changed to remove the errors.

Bit position							
7	6	5	4	3	2	1	0
1	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	0	0	0	0	1
0	1	0	1	1	0	0	0

[2]

MARK SCHEME

5(b)(i)	Bit position								2
	7	6	5	4	3	2	1	0	
	1	0	0	0	1	1	0	0	
	0	0	1	0	0	0	0	0	
	0	0	1	1	0	1	0	1	
	1	1	1	1	0	0	0	1	
	1	1	0	0	0	0	1	0	
	0	0	1	0	0	1	0	0	
	0	0	0	0	0	0	0	1	
	0	1	0	1	1	0	0	0	

9608_s17_qp_13_sc_5.b.ii

QUESTION

(ii) Explain how you arrived at your answers for part (b)(i).

[3]

MARK SCHEME

5(b)(ii)	Three from: Consider each row in sequence Identify any row with incorrect parity Repeat the process for each column in sequence Identify where a row and column with incorrect parity intersect	1 1 1 1	3
----------	---	---	---

QUESTION

- 5 A computer receives data from a remote data logger. Each data block is a group of 8 bytes. A block is made up of seven data bytes and a parity byte.
- Each data byte has a parity bit using odd parity. The parity byte also uses odd parity.
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0	0	1	0	0	0	0	0
1	0	0	1	1	1	0	A
1	1	0	0	0	0	1	0
1	1	0	0	0	0	1	0
1	1	0	0	0	1	1	B
0	0	0	0	0	0	0	0

← Data bytes

0	1	1	0	1	1	0	0
---	---	---	---	---	---	---	---

← Parity byte

- (a) (i) Describe how the data logger calculates the parity bit for each of the bytes in the data block.
-
-
-[2]

MARK SCHEME

5(a)(i)	Count the number of one bits in the <u>first seven</u> bit positions	1	2
	Add a 0 or 1 to bit position 0, to make the count of one bits an <u>odd</u> number	1	

QUESTION

- (ii) State the two missing parity bits labelled A and B.
- A =
- B = [1]

MARK SCHEME

5(a)(ii)	A = 1 B = 1	1
----------	----------------	---

QUESTION

- (iii) Describe how the computer uses the parity byte to perform a further check on the received data bytes.

.....

[2]

MARK SCHEME

5(a)(iii)	Two from: A parity bit is worked out for each <u>column</u> 1 The computer checks the parity of each bit position in parity byte // the computer generates copy of the parity byte and <u>compares</u> 1 If incorrect parity then there is an error in the data received // No parity error means no error in the data received 1 The position of the incorrect bit can be determined 1	2
-----------	---	----------

QUESTION

- (b) (i) A second data block is received as shown in the following table. There are errors in this data block.

Identify and then circle **two** bits in the table which must be changed to remove the errors.

Bit position							
7	6	5	4	3	2	1	0
1	0	0	0	1	1	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	1
1	1	1	1	0	0	0	1
1	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	0	0	0	0	1
0	1	0	1	1	0	0	0

[2]

MARK SCHEME

5(b)(i)	Bit position								2
	7	6	5	4	3	2	1	0	
	1	0	0	0	1	1	0	0	
	0	0	1	0	0	0	0	0	
	0	0	1	1	0	1	0	1	
	1	1	1	1	0	0	0	1	
	1	1	0	0	0	0	1	0	
	0	0	1	0	0	1	0	0	
	0	0	0	0	0	0	0	1	
	0	1	0	1	1	0	0	0	

9608_s17_qp_11_sc_5.b.ii

QUESTION

(ii) Explain how you arrived at your answers for **part (b)(i)**.

[3]

MARK SCHEME

5(b)(ii)	Three from: Consider each row in sequence Identify any row with incorrect parity Repeat the process for each column in sequence Identify where a row and column with incorrect parity intersect	1 1 1 1	3
----------	---	---	---

QUESTION

(b) Data are to be transferred between two devices. Parity checks are carried out on the data.

Explain what is meant by a parity check. Give an example to illustrate your answer.

.....

.....

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.....

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.....

.....

.....[4]

MARK SCHEME

(b) Any **four** from:

- parity can be even or odd
- parity check uses the number of 1s in a binary pattern
- if there is an even / odd number of 1s, then the parity is even / odd
- following transmission ...
- parity of each byte checked
- a parity bit is used to make sure binary pattern has correct parity
- example: 1 0 0 1 0 1 1 1 has parity bit set to 1 in MSB since system uses odd parity (original data: 0 0 1 0 1 1 1 which has four 1 bits)

[4]

QUESTION

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.....[4]

MARK SCHEME

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- parity can be even or odd
- parity check uses the number of 1s in a binary pattern
- if there is an even / odd number of 1s, then the parity is even / odd
- following transmission ...
- parity of each byte checked
- a parity bit is used to make sure binary pattern has correct parity
- example: 1 0 0 1 0 1 1 1 has parity bit set to 1 in MSB since system uses odd parity (original data: 0 0 1 0 1 1 1 which has four 1 bits)

[4]

QUESTION

(c) Data integrity is required at the input stage and also during transfer of the data.

- (i) State **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer.

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..... [3]

MARK SCHEME

(c) (i) **one mark** for each way of maintaining data security + **one mark** for an example/enhancement

- validation (to ensure data is reasonable)
- examples include range checks, type checks, length checks, ...
- verification (checks if data input matches original/if transmitted data matches original)
- can use double data entry or visual check/other methods such as parity checks
- doesn't check whether or not data is reasonable

[3]

QUESTION

- (ii) State **two** ways of maintaining data integrity during data transmission. Use examples to help explain your answer.

.....

.....

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..... [3]

MARK SCHEME

(ii) **one mark** for each way of maintaining data integrity + **one mark** for an example/enhancement

- parity checking
- one of the bits is reserved as parity bit
- e.g. 1 0 1 1 0 1 1 0 uses odd parity
- number of 1s must be odd
- parity is checked at receiver's end
- a change in parity indicates data corruption

- check sum
- adds up bytes in data being sent and sends check sum with the data
- calculation is re-done at receiver's end
- if not the same sum then the data has been corrupted during transmission

[3]

9608_s15_qp_12_sc_4.b

QUESTION

(b) Sensors are used to monitor seismic activity. At the end of each day, all the data are transmitted to a central computer. This is hundreds of kilometres away.

Describe **one** way of ensuring that the integrity of the data is retained during the transmission stage.

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.....[4]

MARK SCHEME

(b) 1 mark for name + 3 marks for description

parity check

- uses even or odd parity which is decided before data sent
- each byte has a parity bit
- parity bit is set to 0 or 1 to make parity for byte correct
- after transmission, parity of each byte re-checked
- if it is different, then an error is flagged
- any reference to use of parity blocks/parity byte to (identify position of incorrect bit)

checksum

- a calculation is carried out on the data to be sent (checksum)
- the result is sent, along with data to recipient
- checksum is re-calculated at receiving end
- if both sums are the same, no error has occurred
- if the sums are different, the data has been corrupted during transmission
- request is sent to re-send data

[4]

9608_s15_qp_11_sc_4.b

QUESTION

- (b) Sensors are used to monitor seismic activity. At the end of each day, all the data are transmitted to a central computer. This is hundreds of kilometres away.

Describe **one** way of ensuring that the integrity of the data is retained during the transmission stage.

.....

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.....[4]

MARK SCHEME

(b) 1 mark for name + **3 marks** for description

parity check

- uses even or odd parity which is decided before data sent
- each byte has a parity bit
- parity bit is set to 0 or 1 to make parity for byte correct
- after transmission, parity of each byte re-checked
- if it is different, then an error is flagged
- any reference to use of parity blocks/parity byte to (identify position of incorrect bit)

checksum

- a calculation is carried out on the data to be sent (checksum)
- the result is sent, along with data to recipient
- checksum is re-calculated at receiving end
- if both sums are the same, no error has occurred
- if the sums are different, the data has been corrupted during transmission
- request is sent to re-send data

[4]